CS211 Test 2 Version 1

Total Mark: 30 marks

1) A segmented address space uses paged virtual memory. Each virtual address has a 2 bit segment number, a 2 bit page number and an 8 bit offset within the page. In the following page tables, a "-" indicates that the page is not in memory. There are a total of 8 page frames.

segment 0		segment 1		segment 2		segment 3	
page	frame	page	frame	page	frame	page	frame
0	-	0	-	0	-	0	-
1	1	1	5	1	3	1	4
2	-	2	6	2	2	2	-
3	0	3	-	3	-	3	7

a) Calculate the total number of bits for a logical address. (1 mark)
ANS: **12 bits**

b) Calculate the total number of bits for a physical address. (1 mark)
ANS: **11 bits**.

c) What is the lowest hexadecimal virtual address in segment 1 that will cause a page fault? (2 marks)
01 00 00000000 = 400_{16}

d) Find the real memory address (in hexadecimal) if the virtual hexadecimal address is B12. (2 marks)
B12 = 1011 0001 0010 → Segment 2 Page 3, -> page fault

2) A register has contents in hexadecimal of BBC01234. After a right **arithmetic** shift of 8 bits, what are the contents of the register? Give your answer in hexadecimal. (2 marks)
Answer:
1011 1011 1100 0000 0001 0010 <u>0011 0100</u> ½ mark

1111 1111 1011 1011 1100 0000 0001 0010 1 mark

=> FFBBC012 ½ mark

3) Justify whether it is possible to design an expanding opcode to allow the following to be encoded in a 16-bit instruction? An address is 4 bits. (5 marks)
15 instructions with three addresses
14 instructions with two addresses
31 instructions with one address
18 instructions with 0 address

Answer:
0000 R1 R2 R3
…. 1 mark
1110 R1 R2 R3

1111 0000 r2 r3

… 1 mark

1111 1101 r2 r3

1111 1110 0000 r3

…. 1 mark

1111 1111 1110 r3

1111 1111 1111 0000

…… 1 mark

1111 1111 1111 1111 => only 16 were possible

Therefore, this design is not possible. 1 mark

{a different justification technique was accepted this time with partial marks but in future, answer this question by designing the expanding opcode}

4) Suppose a CPU has 16 data pins. How many operations will it take to read a 64-bit word. (2 mark)
ANS: 4 operations.

5) Explain whether pipelining reduces the execution time for individual instructions. (2 marks)

No: Each instruction will still take several cycles to pass through the pipeline completely, so the individual instruction execution time is not reduced.

6) A processor uses a fixed-length 32-bit instruction format for a one address machine. The processor has 98 instructions.
   a) Calculate the number of bits for opcode. (1 mark)
   b) Calculate the number of bits for operand. (1 mark)
   c) Calculate the maximum memory address that a program can in this machine can access. (2 marks)
   d) Is it possible to expand the instruction set of this machine? Explain your answer. (2 marks)

Ans:

a) 7 bits
b) 25 bits
c) 33554432 or 2^25
d) Yes. There is space for 30 instructions.

7) Given the following equation:

$$Y = (A + (B \times C)) / (D - E)$$

   a) Write a program to solve the equation using a one-address machine with LOAD, STORE, MUL, ADD, DIV, and SUB instructions. The machine has variable-length instructions. LOAD and STORE instructions have 6 bits of opcode while other instructions have 4 bits of opcode. Operands are of fixed 12 bits. Y,A,B,C,D and E are labels representing the content

of memory locations. A,B,C,D and E must NOT be overwritten. You may use other memory locations (eg. T) as temporary storage. (5 marks)

b) Calculate the length of your program. (2 marks)

ANS: a)

LOAD D

SUB E

STORE T

LOAD B

MUL C

ADD A

DIV T

STORE Y

B) 4 X 18 + 4 X 16 = 136 bits